

Next-generation Bayesian local robustness

Ryan Giordano (rgiordano@berkeley.edu, UC Berkeley)

Berkeley Neyman Seminar

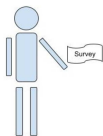
September 2026

The basic problem

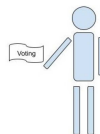
We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.



Observe (x_i, y_i) for $i = 1, \dots, N_S$
(This is the “Data”, $\mathcal{D}$)



Observe x_j for $j = 1, \dots, N_T$

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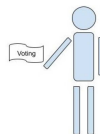
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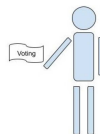
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This talk will study “Multilevel regression and poststratification” estimators:

- Set $\mathbb{E}[y|x, \theta] = m(\theta^\top x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\mathcal{D})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[y|x_j]$.

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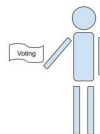
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Problem: MCMC is time consuming and the regression model is surely misspecified.

Goal: Meaningful regression specification checks with frequentist uncertainty.

Result preview

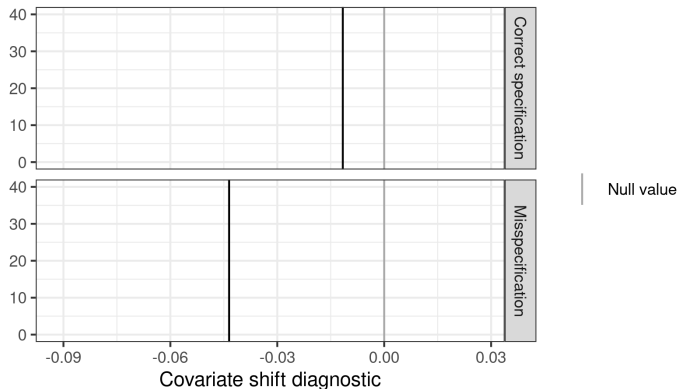


Figure 1: A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecified model is more extreme.

Problem: What is “large enough” to be real evidence of misspecification?

Possible answer: Frequentist confidence intervals via the parametric bootstrap.

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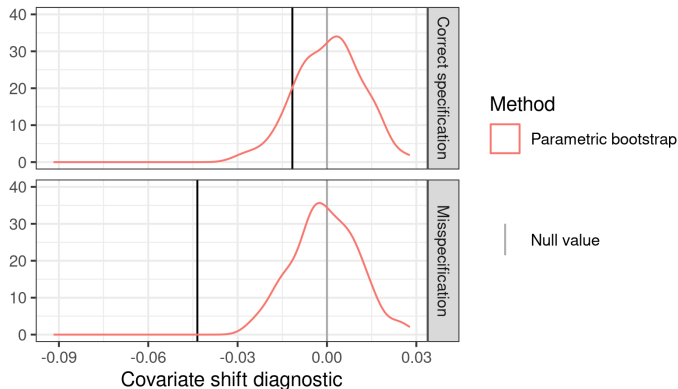


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The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.

Possible answer: Infinitesimal jackknife confidence intervals? [Giordano and Broderick, 2026]

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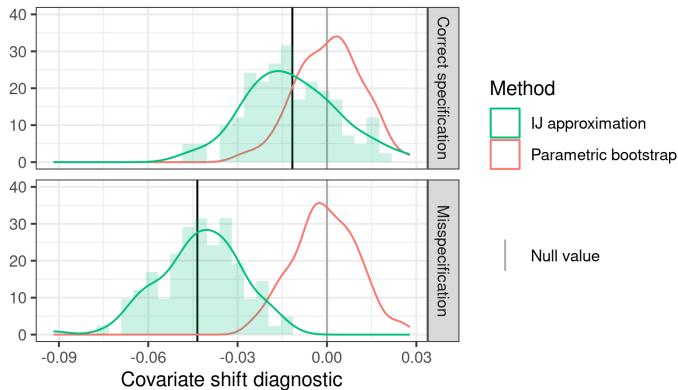


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IJ confidence intervals are fast and often work well for posterior means.
But they are not accurate for this more complex diagnostic (high Monte Carlo error).

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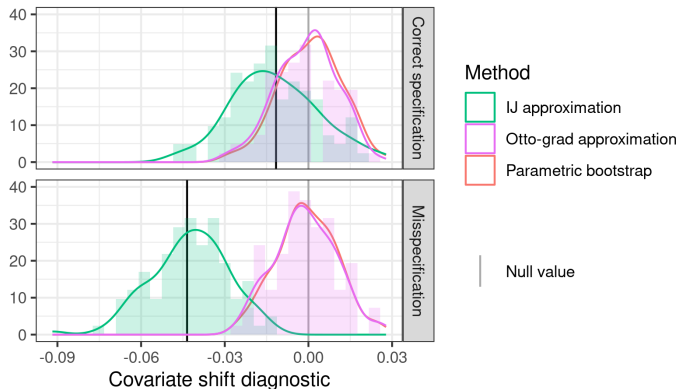


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Answer: Our flow-based local sensitivity measures (“Otto grad”) produce good approximations to the bootstrap at negligible computational expense.

- Local checks of specification of Bayesian regression models
 - (With Alice Cima, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
 - (With Tamara Broderick)
- Bayesian robustness with local flows
 - (With Lucas Schwengber)

TODO: put in pictures

Local checks of specification of Bayesian regression models

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \left[\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j) \right]}_{\text{Want to check specification without re-running MCMC}}.$$

Some existing local approaches:

- Specify an alternative model, maybe test with Bayes factor [Di Noia et al., Cabral et al.]
 - Bayes factors can depend strongly on priors (Lindley's paradox)
 - Articulating a family of models is inconvenient
 - Doesn't take the target into account
- Use out-of-sample checking [Vehtari et al., Kuh et al.]
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The model is definitely misspecified, we want to check whether it's *importantly* misspecified.

Local specification checks

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Let $\mathcal{P}(x, y)$ denote a generic joint density of x, y .

Let $\hat{\mathcal{P}}_S$ the empirical distribution on the survey data.

1. Write our estimand as a functional $\mu(\mathcal{P})$ that takes in a density and returns a MrP estimate
2. Define a covariate density perturbation $w(x)$
3. Define $\mathcal{P}_\epsilon(x, y) = \mathcal{P}(x, y) + \epsilon w(x) \mathcal{P}(y|x) = (\mathcal{P}(x) + \epsilon w(x)) \mathcal{P}(y|x)$
4. Compute the derivative $\Delta(w, \mathcal{P}) = \partial \mu(\mathcal{P}_\epsilon) / \partial \epsilon$
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Lots of similar ideas in causal inference, kernel learning, transfer learning [Shimodaira, Bühlmann, Rojas-Carulla et al., Muandet et al.]. This is just a local Bayesian diagnostics version.

Simple visual example

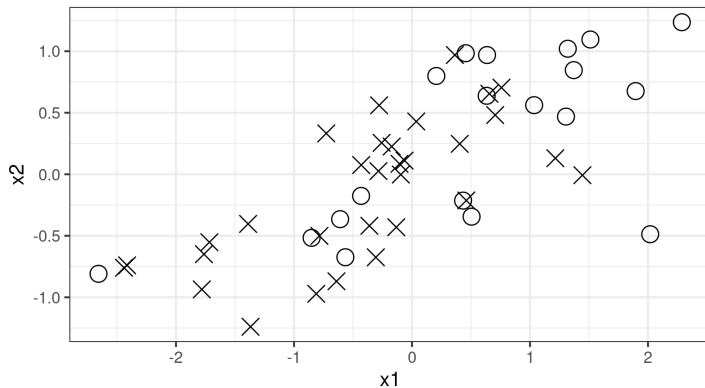


Figure 2: A covariate shift diagnostic

Consider logistic regression with two regressors, x_1 and x_2 .

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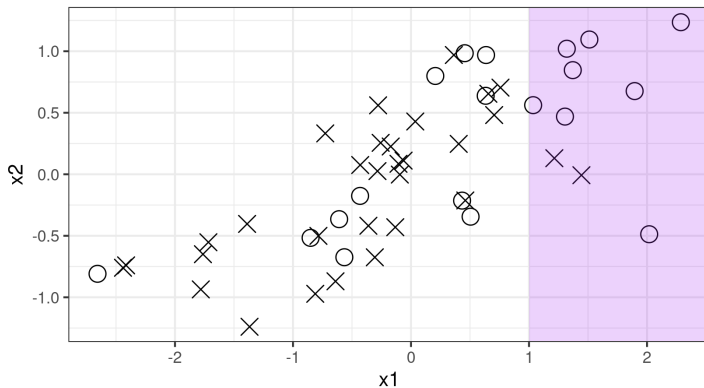


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Taking $w(x) = \mathbb{I}(x_1 > 1)$, we upweight positive examples.

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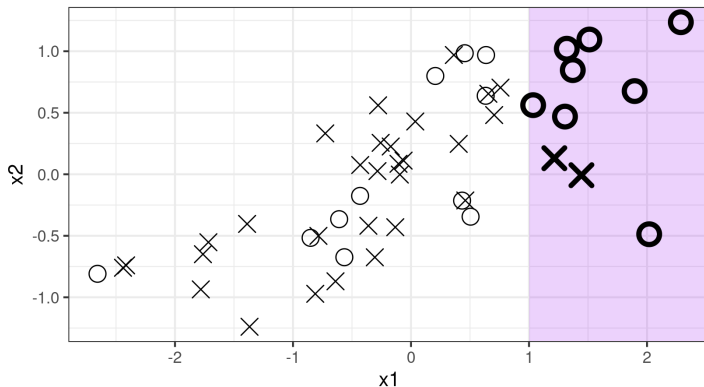


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With enough data, a regression that omits x_1 will change its coefficient of x_2 under re-weighting, whereas a correctly specified regression will not need to.

Bayesian sensitivity analysis

Let the regression log likelihood be $\ell(y|x, \theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i, \theta)\right) \pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i) \ell(y_i|x_i, \theta)\right)$$

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Theorem (Diaconis and Freedman, Gustafson [1996], Giordano et al. [2018]):

Consider a family of distributions $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$ and a function $\phi(\theta)$. By interchange of integration and differentiation,

$$\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) .$$

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Our diagnostic is then given by

$$\Delta(\mathcal{D}) := \left. \frac{\partial \hat{\mu}}{\partial \epsilon} \right|_{\epsilon=0} = \text{Cov}_{\mathcal{P}(\theta|\mathcal{D})} \left(\underbrace{\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\top x_j)}_{\text{Estimand}}, \underbrace{\sum_{i=1}^{N_S} w(x_i) \ell(y_i|x_i, \theta)}_{\text{Log model perturbation}} \right)$$

We expect $\Delta(w, \mathcal{D})$ to be *close* to zero when the model is correctly specified.

Bootstrapping Bayesian sensitivity analysis

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But on any actual dataset, $\Delta(w, \mathcal{D})$ will typically not be precisely zero, even under correct specification.

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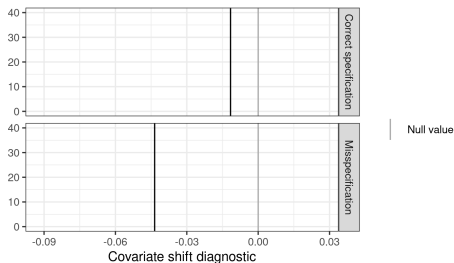


Figure 3: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

Model 0 (correct) : $y \sim \text{Logistic}(1 + X1 + X2 + X3 + X4 + X5)$

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To assess “closeness to zero,” compare with $\Delta(\mathcal{D}^*)$, where $\mathcal{D}^*$ is a bootstrap sample.

- Compute $\Delta(\mathcal{D})$ on the original dataset
- For $b = 1, \dots, 100$:
 1. Sample $\mathcal{D}_b^* := (x_1^*, y_1^*), \dots (x_{N_S}^*, y_{N_S}^*)$ (parametric or nonparametric bootstrap)
 2. Run MCMC to estimate $\mathcal{P}(\theta|\mathcal{D}_b^*)$ (**time consuming!**)
 3. Use the draws to estimate $\Delta(\mathcal{D}_b^*)$
- Compare Δ with distribution of $\Delta(\mathcal{D}_b^*)$.

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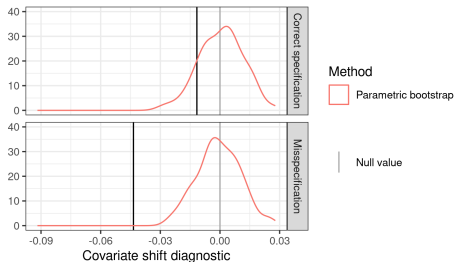


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Frequentist variance with the infinitesimal
jackknife

Why frequentist variance?

For the remainder of the talk, focus on *fast approximations* to the *frequentist* sampling distribution under $x_i, y_i \stackrel{iid}{\sim} \mathcal{P}_S(x, y)$ of Bayesian posterior quantities like our diagnostic.

We will form *local approximations* computable only from *the original posterior*.

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No need to commit to our covariate shift diagnostic to care!

- Frequentist variability of *any* model checking diagnostic is interesting
- Frequentist variability remains meaningful under model misspecification [Huggins and Miller, 2023, Wang et al., 2026, Luo and Ji, 2026, Ji et al., 2025]

We will compare with computationally expensive full bootstrap.

Data re-weighting (again)

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We can form a series approximation:

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If we let $w(x_i)$ denote a *bootstrap weight*, then $\mathcal{P}^\epsilon$ at $\epsilon = 1$ is the bootstrapped posterior!

We can form a series approximation:

$$\begin{aligned} \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D}^*)} [\phi(\theta)] &\approx \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] + \left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} (1-0) \\ &= \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] + \sum_{i=1}^{N_S} \underbrace{\left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial w_i} \right|_{w_i=0}}_{\text{"Empirical influence function"} \frac{\psi_i}{N_S}} w_i \\ &= \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] + \frac{1}{N_S} \sum_{i=1}^{N_S} \psi_i w_i \end{aligned}$$

If this series is accurate, then we can compute ψ_i using the original posterior and quickly approximate bootstrap draws.

Bayesian von-Mises Expansion

Define the “generalized posterior” functional [Giordano and Broderick, 2026]:

$$T(\mathcal{P}, N) := \frac{\int \phi(\theta) \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}{\int \exp \left(N \int \ell(y|x, \theta) \mathcal{P}(x, y) dx dy \right) \pi(\theta) d\theta}.$$

Note that $\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] = T(\hat{\mathcal{P}}_S, N_S)$

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To derive frequentist variance estimates, we study the *von Mises expansion* [von Mises, 1947].

Let $\tilde{\mathcal{P}}^t = t\hat{\mathcal{P}}_S + (1 - t)\mathcal{P}_S$. Then form the series expansion in t :

$$\begin{aligned} \sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) &= \sqrt{N_S} \left(T(\hat{\mathcal{P}}_S, N_S) - T(\mathcal{P}_S, N_S) \right) \\ &= \sqrt{N_S} \left. \frac{\partial T(\tilde{\mathcal{P}}^t, N_S)}{\partial t} \right|_{t=0} (1 - 0) + \mathcal{E}(\tilde{t}) \quad (\tilde{t} \in [0, 1]) \\ &= \sqrt{N_S} \underbrace{\sum_{n=1}^N (\psi_i - \bar{\psi})}_{\text{Infinitesimal jackknife estimator}} + o_p(1) + \mathcal{E}(\tilde{t}). \end{aligned}$$

Bayesian von-Mises Expansion Results

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) = \sqrt{N_S} \sum_{n=1}^N (\psi_i - \bar{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

Theorem 3 [Giordano and Broderick, 2026] (sketch):

Under Bernstein-von Mises-like conditions, $\sup_{\tilde{t} \in [0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{P} 0$.

Corollary [von Mises, 1947]: Let $\frac{1}{N_S} \sum_{i=1}^{N_S} (\psi_i - \bar{\psi})^2 \xrightarrow{P} V$. By the CLT applied to ψ_i ,

$$\sqrt{N_S} \left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} [\phi(\theta)] - T(\mathcal{P}_S, N_S) \right) \xrightarrow{d} \mathcal{N}(0, V).$$

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To compute and study the parametric bootstrap instead is a simple change:

$$\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp \left(\sum_{i=1}^{N_S} \epsilon (\ell(y_i^*|x_i, \theta) - \ell(y_i|x_i, \theta)) \right).$$

(See, e.g., Giordano et al. [2026] Theorem 4.1).

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Note: The von-Mises expansion is stronger than necessary for consistency [Giordano and Broderick, 2026]. But it can be used to prove *lower bounds* on $\mathcal{E}(\tilde{t})$ and so *inconsistency* when no BVM theorem holds.

Data re-weighting.

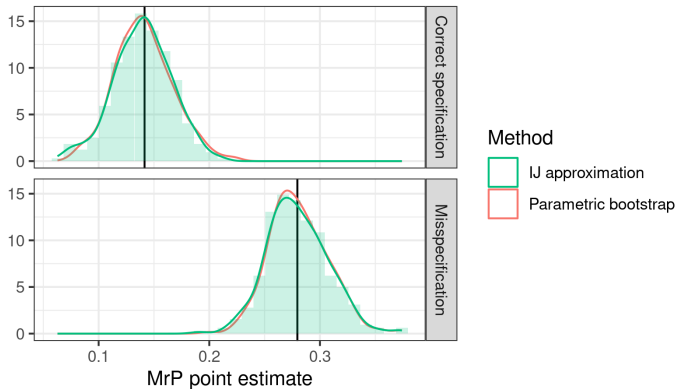


Figure 4: MrP Posterior Mean

The IJ does great for the MrP point estimate. See also Luo and Ji [2026], Ji et al. [2025].

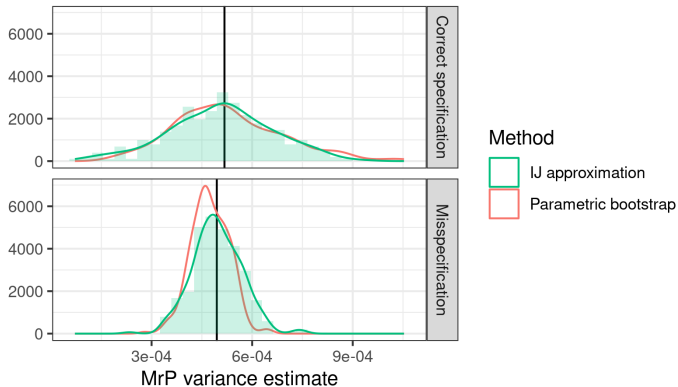


Figure 4: MrP Posterior Variance

The IJ does pretty well for the MrP posterior variance.

Data re-weighting.

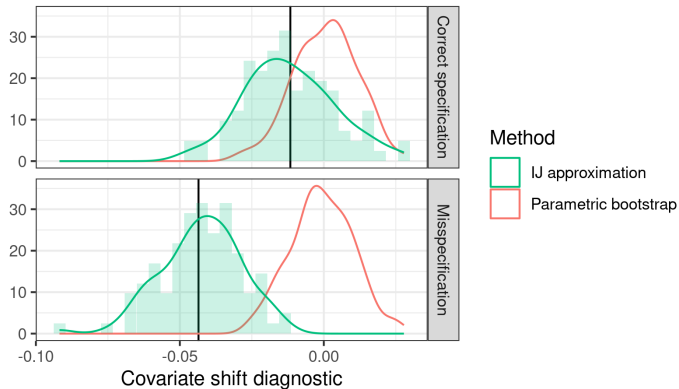


Figure 4: Covariate Shift Diagnostic

Need something better than the IJ for noisier / more complex posterior quantities!

Bayesian robustness with local flows

Why flows for sensitivity analysis?

Current Bayesian local robustness is built around linearizing posterior expectations:

$$\mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)] \approx \mathbb{E}_{\mathcal{P}(\theta)} [\phi(\theta)] + \left. \frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)} [\phi(\theta)]}{\partial \epsilon} \right|_{\epsilon=0} (\epsilon - 0),$$

where the derivative is estimated — with noise — from MCMC draws [Gustafson, 1996, Giordano et al., 2018, Cabral et al., Di Noia et al., Giordano et al., 2026].

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This has some shortcomings:

- Potential MCMC variance, not obvious how to smooth
- Linear extrapolation can lead to impossible values (e.g. negative variances)
- Complex quantities like posterior quantiles are hard to linearize
- Bayesians like to work with draws, not analytic expectations [Han et al., 2025]

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Idea: Instead of linearizing expectations, learn a flow from $\mathcal{P}$ to $\mathcal{P}^\epsilon$.

There's a large literature on learning flows between distant distributions. [Papamakarios et al., 2021]

Our problem is easier:

- Only need “local” flows
- Our perturbations have a lot of structure (e.g. bootstraps)

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

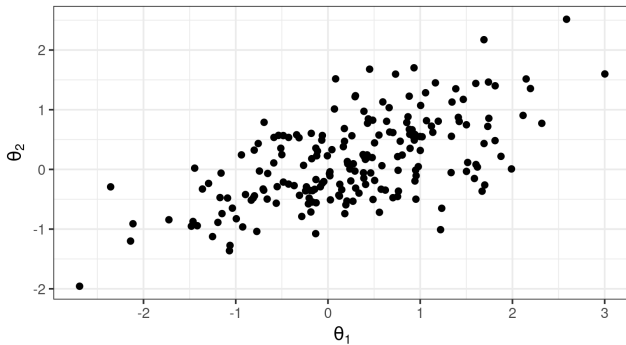


Figure 5: Draws from a posterior

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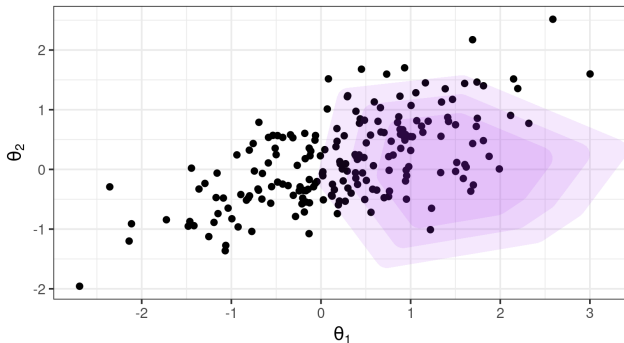


Figure 5: A posterior perturbation

Imagine the density $\mathcal{P}(\theta)$ coming out of the page. The perturbation $h(\theta)$ is defined “vertically.”

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

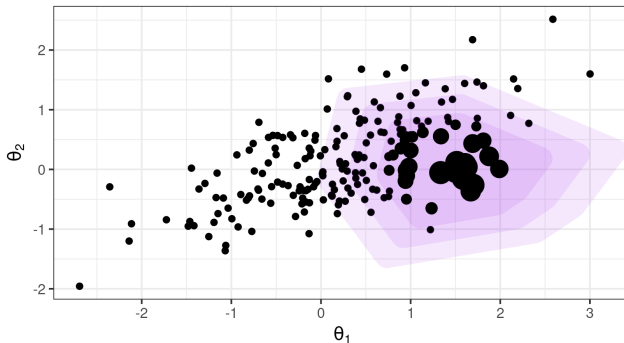


Figure 5: A posterior perturbation with weights

Plug-in estimates for the covariance are equivalent to producing the perturbation by weighting (Appendix B Giordano et al. [2018])

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$, with draws $\theta \sim \mathcal{P}(\theta)$.

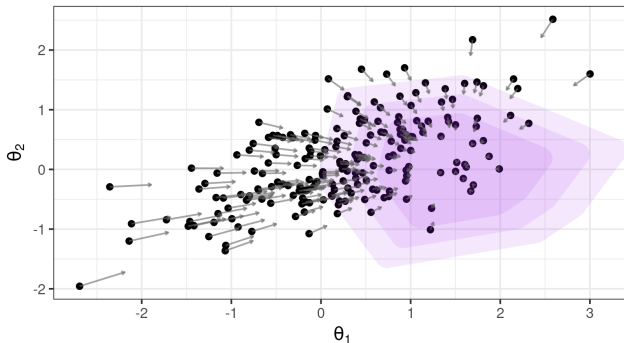


Figure 5: A posterior perturbation with a flow

We will try, instead, try to effect the perturbation with a flow $\theta \mapsto \Phi^\epsilon(\theta)$

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta) \exp(\epsilon h(\theta))$.

We want a flow $\theta \mapsto \Phi^\epsilon(\theta)$ such that, for small ϵ , $\Phi^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $\Phi^0(\theta) = \theta \sim \mathcal{P}(\theta)$.

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Suppose there is a scalar-valued potential function $\phi(\theta)$ such that

$$\Phi^\epsilon(\theta) = \theta + \epsilon \nabla \phi(\theta) + O(\epsilon^2) \quad \text{as } \epsilon \rightarrow 0. \text{ Then:}$$

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$$(\star) \quad \text{Cov}_{\mathcal{P}(\theta)} (\phi(\theta), h(\theta)) = \mathbb{E}_{\mathcal{P}(\theta)} [\nabla \phi(\theta)^\top \nabla \phi(\theta)] \quad \text{(Connecting } h \text{ and } \phi)$$

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Theorem (Johnsen, Kipnis and Varadhan, Bakry et al. [2014], Schwengber 2026):

Assume that $\mathbb{E}_{\mathcal{P}(\theta)} [h(\theta)^2] < \infty$ and $\mathcal{P}(\theta)$ has sub-Gaussian tails. Then there exists a solution ϕ to the differential equation

$$\mathcal{L}_{\mathcal{P}}(\phi) := -\nabla \phi^\top \nabla \log \mathcal{P} - \Delta \phi = h$$

such that $\|\nabla \phi\|_{L^2(\mathcal{P})} < \infty$ and the flow $\theta_\epsilon = \theta + \epsilon \nabla \phi(\theta)$ satisfies Equation $(\star)$.

Note that $\mathcal{L}_{\mathcal{P}}$ is well-known: it's the drift operator, generator of a Langevin diffusion, and a Stein operator.[Ambrosio et al., 2021, Song et al., 2021, Ross, 2011]

How can we find $\phi(\theta)$ in practice?

Solving PDEs is hard. How can we easily find $\phi(\theta)$ in practice?

$$(\star) \quad \text{Cov}_{\mathcal{P}(\theta)}(\phi(\theta), h(\theta)) = \mathbb{E}_{\mathcal{P}(\theta)}[\nabla \phi(\theta)^\top \nabla \phi(\theta)] \quad (\text{Identity connecting } h \text{ and } \phi)$$

Basis expansion approach (the Galerkin method [Ern and Guermond, 2004]):

- Specify a family of real-valued target functions $\phi_n(\theta)$ for $n \in [N]$
- Specify a family of real-valued basis functions $b_m(\theta)$ for $m \in [M]$

Let $\phi(\theta; \beta) := \sum_{m=1}^M \beta_m b_m(\theta)$ and plug into Equation $(\star)$.

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Stacking in matrices:

$$\underbrace{\begin{bmatrix} \vdots \\ \text{Cov}_{\mathcal{P}(\theta)}(\phi_n(\theta), h(\theta)) \\ \vdots \end{bmatrix}}_{N \times 1 \text{ matrix}} \underbrace{\begin{bmatrix} \vdots \\ \dots \quad \mathbb{E}_{\mathcal{P}(\theta)}[\nabla \phi_n(\theta)^\top \nabla b_m(\theta)] \quad \dots \\ \vdots \end{bmatrix}}_{N \times M \text{ matrix}} = \begin{bmatrix} \vdots \\ \beta_m \\ \vdots \end{bmatrix}$$

This gives N linear equations in M unknowns, and the matrices can be estimated with MCMC.

Theorem (Gohberg and Kreĭn [1969], Bakry et al. [2014], Schwengber 2026): When $\mathcal{P}(\theta)$ is Gaussian, taking $N = M$ and $\phi_n(\theta) = b_n(\theta)$ to be the Hermite polynomials give the best finite-rank approximation to $\mathcal{L}_{\mathcal{P}}^{-1}$ in its operator norm.

Using the 1^{st} order Hermite polynomials $\phi_n(\theta) = b_n(\theta) = \theta_m$, the estimated flow is given by

$$\theta_{\epsilon} = \theta + \epsilon \underbrace{\text{Cov}_{\mathcal{P}(\theta|\mathcal{D})}(\theta, h(\theta))}_{\left. \frac{\partial \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[\theta]}{\partial \epsilon} \right|_{\epsilon=0}}$$

This just moves all points in the direction of the mean shift!

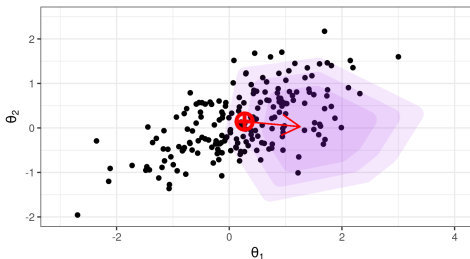
An elegant flow

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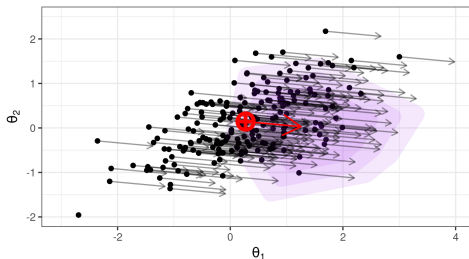
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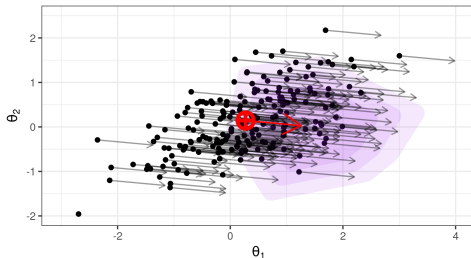
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This just moves all points in the direction of the mean shift!



Suppose we have $\theta_m \sim \mathcal{P}(\theta|\mathcal{D})$, and bootstrap weights w_i .

Set $\theta_m^* = \theta_m + \frac{1}{N_S} \sum_{i=1}^{N_S} w_i \psi_i$ and compute any posterior quantities from these samples.

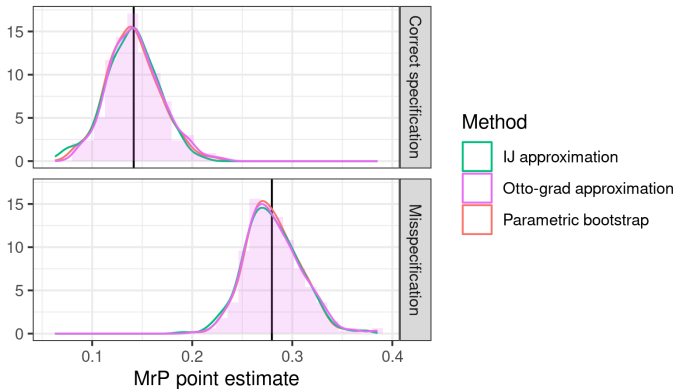


Figure 6: MrP Posterior Mean

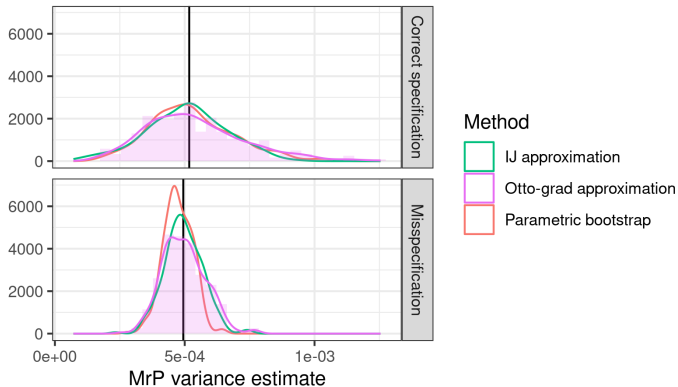


Figure 6: MrP Posterior Variance

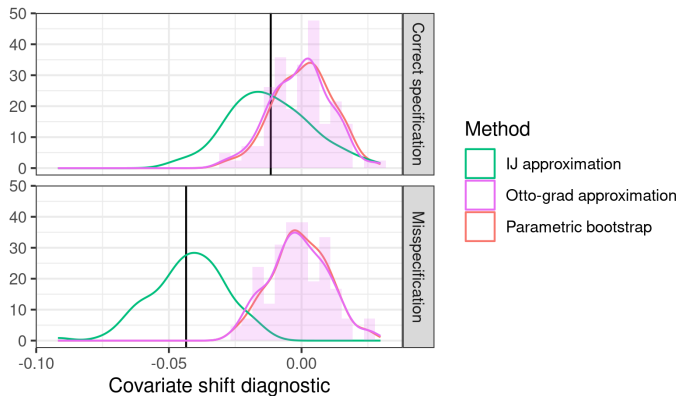


Figure 6: Covariate Shift Diagnostic

Carefully understanding the gap between IJ and the flow is ongoing work!

- Local checks of specification of Bayesian regression models
 - Use invariance to covariate shift as a diagnostic for Bayesian hierarchical models
- Frequentist variability of Bayesian posteriors with the infinitesimal jackknife
 - Use sensitivity of mean parameters to speed up bootstrapping MCMC
- Bayesian robustness with local flows
 - Use easy-to-compute approximate flows to speed up bootstrapping MCMC

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